

EECE 571Q

Lecture 02: Don't worry, we can fix it

Thursday 14 May 2026

Announcements

- Quiz 1 Tuesday - sign up for PrairieLearn, bring a laptop
- Supplementary exercises available on PrairieLearn
- Suggested reading available at end of slide decks

Last time

	Pure state	Pure state ρ	Mixed state ρ
States	$ \psi\rangle$	$\rho = \psi\rangle\langle\psi $	$\rho = \sum_i p_i \psi_i\rangle\langle\psi_i $
Ops.	$ \psi'\rangle = U \psi\rangle$	$\rho' = U\rho U^\dagger$	$\rho' = \sum_i p_i U \psi_i\rangle\langle\psi_i U^\dagger$
Meas. $\{ \psi_i\rangle\}$	$ \langle\psi_i \psi\rangle ^2$ $\langle B \rangle = \langle \psi B \psi \rangle$	$\text{Tr}(\psi_i\rangle\langle\psi_i \rho)$ $\text{Tr}(B\rho)$	$\text{Tr}(P_i \rho)$ $\text{Tr}(B\rho)$

$$\sum_i p_i = 1$$

Last time

We expressed operations, measurement, and errors as **quantum channels** in the Kraus representation.

$$\Phi(\rho) = \sum_i K_i \rho K_i^\dagger \quad \sum_i K_i^\dagger K_i = I$$

$$\mathcal{U}(\rho) = U \rho U^\dagger$$

$$\Phi_M(\rho) = \sum_i \text{Prob}(i) \frac{K_i \rho K_i^\dagger}{\text{Prob}(i)}$$

$$\Phi_{BF}(\rho) = (1-p)\rho + p \cdot X \rho X$$

$$E_0 = \sqrt{1-p} I$$

$$E_1 = \sqrt{p} X$$

Learning outcomes

- 1 Describe, and experimentally estimate, processor quality metrics
- 2 Define a quantum error correcting code and its three key parameter n, k, d .
- 3 Design circuits to detect and correct bit flip, phase flip, and arbitrary errors
- 4 Outline the conditions under which errors can be corrected

How bad is it?

One approach for quantifying the strength and effects of noise is to define mathematical distance functions between density matrices.

A common metric is the **trace distance**:

$$T(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1 = \frac{1}{2} \text{Tr} \left(\sqrt{(\rho - \sigma)^\dagger (\rho - \sigma)} \right)$$

Bounded by $0 \leq T(\rho, \sigma) \leq 1$; *lower* trace distance is better.

How bad is it?

Can also compare using **fidelity**,

$$F(\rho, \sigma) = \left(\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right)^2$$

Bounded by $0 \leq F(\rho, \sigma) \leq 1$. *Higher* fidelity is better.

Special case: if ρ is pure but σ is mixed,

$$\rho = |\psi\rangle\langle\psi|$$

$$F(\rho, \sigma) = \langle \psi | \sigma | \psi \rangle$$

Practice exercise: derive this and other special cases (see PrairieLearn).

How bad can it get?

Exercise: Determine $F(\rho, \sigma)$ if ρ is pure and $\sigma = \Phi_{BF}(\rho)$ with parameter p . $\rho = |\psi\rangle\langle\psi| = \langle\psi|\sigma|\psi\rangle = (1-p)\rho + p X\rho X$

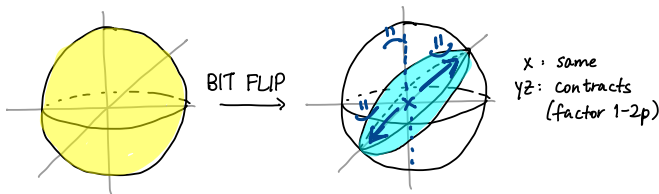
$$\begin{aligned} F(\rho, \sigma) &= \langle\psi|\Phi_{BF}(\rho)|\psi\rangle \\ &= 1 - p + p \langle X \rangle^2 \\ &\geq 1 - p \end{aligned}$$

$$\downarrow$$

$$\langle\psi|X|\psi\rangle$$

Best case: $|+\rangle/|-\rangle$
= 1

Worst case: $|0\rangle/|1\rangle$
 $f(p)$



How bad can it get?

Exercise: What happens if the strength of depolarizing noise is $p = 3/4$? What is the fidelity of any pure ρ with the output of such a channel?

$\lambda = 1$

$$\Phi_b(\rho) = (1-p)\rho + \frac{p}{3}X\rho X + \frac{p}{3}Y\rho Y + \frac{p}{3}Z\rho Z$$

$$\Phi_b(\rho) = (1-\lambda)\rho + \lambda \frac{I}{2}$$

$$\Downarrow$$

$$\Phi_b(\rho) \stackrel{\lambda=1}{=} \frac{I}{2} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix}$$

$$F(|\psi\rangle\langle\psi|, \Phi_b(\rho)) = \langle\psi|\Phi_b(\rho)|\psi\rangle = \frac{1}{2}$$

Amplitude damping channel

Suppose $|1\rangle$ relaxes to $|0\rangle$ with probability p .

Doesn't relax:

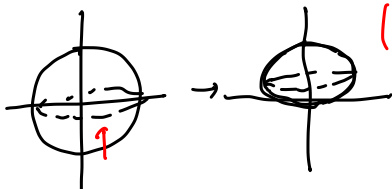
$$\begin{array}{l} |1\rangle\langle 1| \\ |0\rangle\langle 0| \end{array} \Rightarrow K_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot f(p)$$

Does relax:

$$|0\rangle\langle 1| = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \Rightarrow K_1 = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix}$$

How do you think this affects the Bloch sphere?

$$K_0^\dagger K_0 + K_1^\dagger K_1 \neq I$$



$$\begin{pmatrix} 1 & 0 \\ 0 & 1-p \end{pmatrix} \quad \begin{pmatrix} 0 & 0 \\ 0 & p \end{pmatrix} \Rightarrow K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix} \quad K_1 = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix}$$

Amplitude damping channel

Amplitude damping is related to a different quality metric.

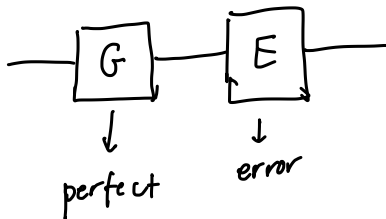
Assume probability of remaining in $|1\rangle$ decays with time,

$$p(t) = e^{-t/\tau}$$

Let's do some experiments to simulate qubit characterization.

Quantum error correction

Our current picture of noise:



We can protect qubits against noise using error correcting codes.

Repetition codes

Imagine a bit string transmitted through a classical channel that flips each bit (individually) with probability p .

Idea: add redundant information to enable *detection* and *correction* of bit flips.

Define *encoding* and *decoding* operations, $\mathcal{U}$ and $\mathcal{D}$,

Repetition codes

Devise a procedure, $\mathcal{R}$, to recover from an error: majority voting.

Operation	Outcome
	0 0 1 0 1
U	000 000 111 ...
E	010 000 101 ...
$\mathcal{R}$	000 000 111
D	0 0 1

error prob = p

Is this better? When will this fail? $\rightarrow$ 2 bit flips

$$\text{prob 2 bit flips: } 3p^2(1-p) > \text{sum } 3p^2 - 2p^3 = p_e$$

$$\text{prob 3 bit flips: } p^3$$

$$p_e < p \Rightarrow \checkmark$$

Quantum repetition code (bit flip code)

Idea: $U(|\psi\rangle) = |\psi\rangle^{\otimes 3}$

Why won't this work?

- No cloning
- Phase errors — other errors, coherent errors
- Destructive meas.

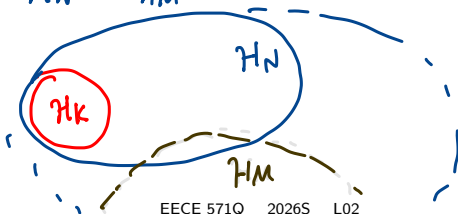
Definition: quantum error correcting code (QECC)

Source: Gottesman, Def. 2.1

"A quantum error-correcting code $(U, \mathcal{E})$ (or QECC for short) is a **partial isometry** $U : \mathcal{H}_K \rightarrow \mathcal{H}_N$ with the set of **correctable errors** $\mathcal{E}$ (consisting of linear maps $E : \mathcal{H}_N \rightarrow \mathcal{H}_M$) with the following property: $\exists$ quantum operation $\mathcal{D} : \mathcal{H}_M \rightarrow \mathcal{H}_K$ such that $\forall E \in \mathcal{E}, \forall |\psi\rangle \in \mathcal{H}_K$,"

$$\mathcal{D}(EU|\psi\rangle \langle\psi| U^\dagger E^\dagger) = c(E, |\psi\rangle) |\psi\rangle \langle\psi|$$

code : maps $\mathcal{H}_K \rightarrow \mathcal{H}_N$ (logical space $\rightarrow$ physical space)
errors: map $\mathcal{H}_N \rightarrow \mathcal{H}_M$



Bit flip code: encoding circuit

Alternative:

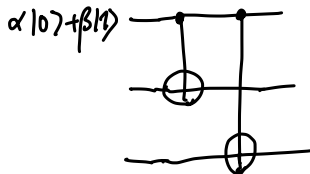
$$U(|0\rangle) = |000\rangle = |\bar{0}\rangle = |0\rangle_L$$

$$U(|1\rangle) = |111\rangle = |\bar{1}\rangle = |1\rangle_L$$

Exercise:

- (a) What is the codespace? $\rightarrow$ 2D subspace of $\mathcal{H}_{2^3}$
 (b) Design a circuit that implements U

$$U(\alpha|0\rangle + \beta|1\rangle) \mapsto \alpha|000\rangle + \beta|111\rangle$$



Designing logical operations

Logical operations are specific to codes.

They should act on the logical states $|0\rangle_L$, $|1\rangle_L$ the same way the physical operations act.

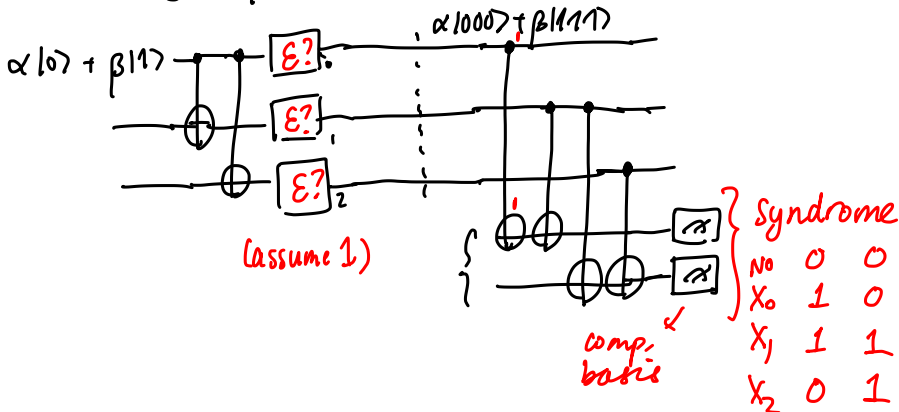
Exercise: design circuits for logical X , Z , H , S , and $CNOT$.

$$\begin{aligned}\bar{X}_L |0\rangle &= |1\rangle \\ \bar{X}_L |1\rangle &= |0\rangle \\ &\vdots\end{aligned}$$

Bit flip code: recovery operation

Let's design a circuit to detect (a) *whether* an error occurs, and
(b) *which* error occurs.

$$\mathcal{E} = \{XII, IXI, IIX\}$$



Bit flip code: recovery operation

Design a circuit to *recover* the state of the first qubit

conditional ops. based on syndrome.

Bit flip code: full circuit for bit flip code

Bit flip code: analysis

Compare overall fidelity of the state *after the error*, vs. fidelity *after recovery*.

After the error (1 qubit, bit flip error)

$$\Phi(\rho) = (1-p) |\psi\rangle\langle\psi| + p X |\psi\rangle\langle\psi|$$

$$F(|\psi\rangle\langle\psi|, \Phi(\rho)) \geq 1-p$$

encoding, error, then

After recovery, have the state

$$F(|\psi\rangle, \mathcal{R}(\Phi^{\otimes 3}(\mathcal{U}(|\psi\rangle\langle\psi|)))) \geq (1-p)^3 + 3p(1-p)^2$$

Phase flip errors

We will start here Tuesday.

With our encoding

and appropriate circuitry, we can correct (at most) *one bit flip error*, but no phase flip errors.

Phase flip code: encoding circuit

Main idea: make phase flip errors “look like” bit flip errors.

Conditions for quantum error correction

How do we know if a particular error is correctable by a QECC?

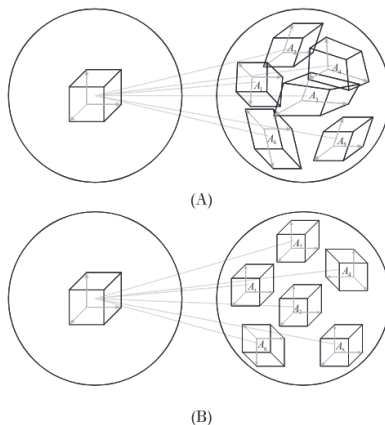


Image: Nielsen and Chuang

Conditions for quantum error correction

Sufficient conditions: non-degenerate code; errors map to orthogonal subspaces.

Formal statement:

“($Q, \mathcal{E}$) is a QECC iff $\forall |\psi\rangle, |\phi\rangle \in Q, \forall E_a, E_b \in \mathcal{E}$,

Note that C_{ab} does not depend on $|\psi\rangle$ or $|\phi\rangle$ ”

Source: Gottesman, Thm. 2.7

Correcting arbitrary errors

Fact: QECCs are linear.

$$\mathcal{D}(EU|\psi\rangle\langle\psi|U^\dagger E^\dagger) = c(E, |\psi\rangle)\langle\psi|$$

If E_1 and E_2 are correctable, then

$E_1 E_2$ is correctable. (Supplementary exercise; see Gottesman 2.4)

Consequence: if we can correct Pauli errors $\mathcal{E} = \{I, X, Z, XZ\}$, we can correct *arbitrary* errors.

Correcting arbitrary errors

Bit flip, phase flip: obviously Paulis

Coherent errors?, e.g., slight over-rotation by angle θ .

Amplitude damping? *See supplementary exercises*

Shor code

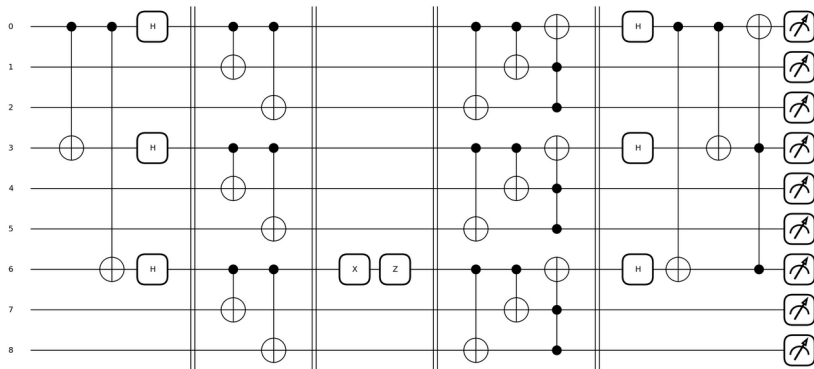
To correct a combination of one bit flip and/or phase flip error, we can *concatenate* codes.

Shor code

The Shor code can correct one *arbitrary error on a single qubit*.

Imagine a bit + phase flip on qubit 6:

Shor code



What about errors on *multiple* qubits?

Source: Gottesman 1.3.2-1.3.4 (see 1.3.4 for proofs)

One option: model as *independent channels*, i.e.,

Revisit our bit flip channel...

If we had $\Phi_{BF} \otimes \Phi_{BF} \otimes \Phi_{BF}$ and p is small, many terms contribute very little.

Often more convenient to model using t -qubit error channels.

What about errors on *multiple* qubits?

Source: Gottesman 1.3.3 - see section for proofs

A linear operator has **weight** t if it acts non-trivially on t qubits tensored with identity on all other qubits.

A linear operator is a **t -qubit error** if it has the form

and each E_i has weight at most t (not necessarily on same qubits).

An n -qubit channel is a **t -qubit error channel** if it has a Kraus decomposition where all Kraus operators are t -qubit errors.

Code distance

The *distance* of a code is the minimum weight, d , of an error F such that the QECC conditions are violated, i.e.,

(for all possible pairs of codewords).

If E_a , E_b are weight t errors, F has weight at most $2t$.

Consequence: a distance d code

- corrects up to $\lfloor (d-1)/2 \rfloor$ errors
- detects up to $d-1$ errors

Code distance

The Shor code is an

$$((n, k, d)) = ((9, 1, 3))$$

code.

Example: find a minimum-weight F error for 9-qubit code that doesn't satisfy criteria.

Next time

Next/ class:

- Stabilizer codes

Action items:

- ① Prepare for quiz 1
- ② Sign up for PrairieLearn
- ③ Start thinking about project topics
- ④ Supplementary exercises

Recommended reading:

- From this class: Gottesman Ch. 1.3, 2; Codebook DM, EC; Nielsen and Chuang 8.1-8.3, 9, 10.1-10.3
- For next class: Gottesman Ch. 3, Codebook EC; N&C 10.5